

Arjuna JEE 2025

Maths

DPP: 3

Basic Mathematics

Q1 Given $(x^2 + 7x \geq 8)$ is called as _____.

- (A) Equation (B) Inequation
(C) Polynomial (D) None of these

Q2 If we multiply whole inequality by a negative number then

- (A) sign of inequality remains same.
(B) sign of inequality gets reversed.
(C) depends, which negative number is multiplied.
(D) depends, both side of inequality are positive or negative.

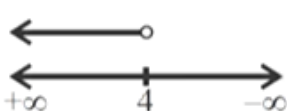
Q3 "Zero" is

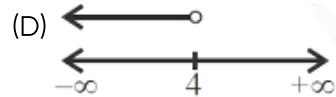
- (A) positive number
(B) negative number
(C) both positive as well as negative
(D) neither positive nor negative

Q4 If $\alpha \geq \beta$ then

- (A) $2\alpha \geq 3\beta$
(B) $\alpha \leq -\beta$
(C) $\alpha^2 > \beta^2$
(D) $-\alpha \leq -\beta$

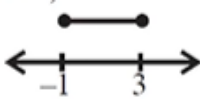
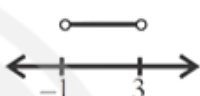
Q5 Correct number line representation for: $(x < 4)$

- (A) 
(B) 
(C) 



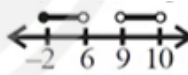
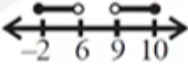
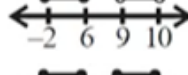
Q6 Correct number line representation for:

$$(-1 < x \leq 3)$$

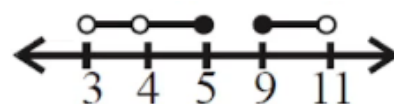
- (A) 
(B) 
(C) 
(D) 

Q7 Correct number line represents for:

$$x \in [-2, 6) \cup (9, 10]$$

- (A) 
(B) 
(C) 
(D) 

Q8 Correct option represents



- (A) $x \in (3, 5] \cup [9, 11)$
(B) $x \in (3, 5] \cup [9, 11) - \{4\}$



(C) $x \in (3, 4) \cup (4, 5) \cup [3, 11)$

(D) $x \in (3, 4) \cup (4, 5] \cup (9, 11)$

Q9 If α & β lies on opposite sides of "zero" on number line then

(A) $\alpha > \beta$

(B) $\beta > \alpha$

(C) $\alpha\beta < 0$

(D) $\alpha\beta > 0$

Q10 Solution of inequality: $\frac{x-8}{2} \leq 5$

(A) $x < 2$

(B) $x \geq 18$

(C) $x \leq 18$

(D) $x \geq 2$

Q11 Solution of inequality $2x - 3 > 0$ and $4 - 3x < 0$.

(A) $x > \frac{4}{3}$

(B) $x < \frac{4}{3}$

(C) $x > \frac{3}{2}$

(D) $x < \frac{3}{2}$

Q12 Solution of inequality $-5 < x + 3 \leq 9$

(A) $x \in [-8, 6)$

(B) $x \in (-8, 6]$

(C) $x \in (-2, 6]$

(D) $x \in (-8, 6)$

Q13 Identify the solution set for

$-(x - 3) < (5 - 2x)$

(A) $(-\infty, -1)$

(B) $(-\infty, -5)$

(C) $(-\infty, 2)$

(D) None of these

Q14 What is the solution set of $\frac{2(x-1)}{5} \leq \frac{3(2+x)}{7}$

(A) $(-4, \infty)$

(B) $(24, \infty)$

(C) $(-12, \infty)$

(D) $[-44, \infty)$

Q15 Identify the solution set of

$\frac{(x-1)}{3} + 4 < \frac{x-5}{5} - 2$

(A) $(-\infty, -5)$

(B) $(-\infty, -50)$

(C) $(-\infty, -15)$

(D) $(-\infty, -10)$

Q16 What is the solution set of $0 < \frac{-x}{2} < 3$

(A) $(5, 6)$

(B) $(-6, 0)$

(C) $(-6, 6)$

(D) None of these

Q17 Solution set of the inequality $\frac{3(x-2)}{5} \geq \frac{5(2-x)}{3}$ is

(A) $(-\infty, 2)$

(B) $(-\infty, 2]$

(C) $[2, \infty)$

(D) $(2, \infty)$



Answer Key

Q1 (B)
Q2 (B)
Q3 (D)
Q4 (D)
Q5 (D)
Q6 (D)
Q7 (B)
Q8 (B)
Q9 (C)

Q10 (C)
Q11 (C)
Q12 (B)
Q13 (C)
Q14 (D)
Q15 (B)
Q16 (B)
Q17 (C)



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